Shivam Kataria

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Education

University of British Columbia May 2026

Bachelor of Applied Science - Mechanical Engineering, Cumulative Average: 81%, Dean's Honour List

Relevant Courses: Real-Time Embedded Systems, Electronics, Power Electronics, Control Systems, Computer Architecture, OS

Experience

Bosch Sep 2024 – Aug 2025

Embedded Systems Engineering Intern *Renningen, Germany*

- Conducted R&D for a fast PCR device - driving advancements in diagnostic capabilities for biomedical applications
- Designed and implemented **real-time I2C** driver for power monitoring in **C** using **STM32** and **RTOS**, employing high-speed I2C and **DMA**, optimizing for efficient embedded software in a complex and time-critical system
- Contributed to PCB design in **Altium** by creating and updating schematics, creating custom component footprints, etc.
- Debugged **mixed-signal PCBA** and FreeRTOS embedded firmware using tools such as **logic analyzer**, **oscilloscope**, **DMM**, and **SWD** to identify and resolve issues across digital, analog, and power domains
- Implemented and tuned **PID** and sliding mode control (**SMC**) algorithms in a real-time embedded system on STM32, using **ADC**-based temperature feedback to achieve **2x** faster thermal cycling and minimizing tracking error
- Automated fast thermal cycling experiments with Peltier modules using **Python** and source measure unit (SMU) via the VISA protocol to capture precise thermal data using thermocouples, increasing experimentation speeds by **50%**
- Analyzed experimental data using Python, **Pandas** and, Matplotlib to extract thermal performance insights, enabling **data-driven design optimizations** for enhanced cycling efficiency and speed to meet design criteria

Cibotica Sep 2023 – Dec 2023

Embedded Software Developer Intern *Vancouver, Canada*

- Developed software contributing to the **successful first launch** of the early food tech startup in a **fast-paced** and **agile product-focused** environment using **Raspberry Pi** and **Pico MCUs**, improving the product's speed by **80%**
- Built and debugged state machine structured firmware for 16 components using CircuitPython consisting of several motors, sensors and peripherals such as ADC, I2C, SPI, GPIO, etc. for precision dispensing control
- Constructed embedded **CAN** protocol software using CircuitPython for updating machine parameters, leading to **2x** speed
- Implemented a **ROS2** node for simulating a salad-making system, optimizing testing and debugging processes by **90%**
- Automated ROS2 topics analyzing for system monitoring and push notification events to the backend using WebSockets
- Containerized ROS2 nodes for enhanced portability and scalability using **Docker**, increasing deployment speed by **80%**
- Validated and troubleshot PCBAs using oscilloscope and DMM for comprehensive analysis and diagnostics

Defence Research and Development Canada (DRDC) Jan 2023 – Aug 2023

Robotics Researcher Intern *Canadian Forces Base Suffield, Canada*

- Developed and improved drone tracking algorithms using deep reinforcement learning (RL) and PID controller to track and identify target drone in real-time using YOLO object detection algorithms and RL library StableBaselines3
- Increased tracking speed with RL in Unreal Engine simulations by **3x** through hyperparameters and reward function tuning
- Tuned PID tracking controller to minimize overshoot and instability, increasing tracking speed by **5x** in real-flight operations
- Trained YOLO algorithm on a custom data set of real flight images to improve detection speed and accuracy on drones
- Modeled drone onboard computer mounting hardware in SolidWorks and produced detailed engineering drawings to verify dimensions and tolerances prior to 3D printing

Engineering Student Teams

UBCO Aerospace Jan 2022 – May 2024

Vice President of Software *Kelowna, Canada*

- Placed **2nd** with the **Innovation Award** in the Unmanned Systems Canada Student UAS Competition in 2022 and 2023
- Programmed an unmanned aerial vehicle (UAV) to complete mission requirements autonomously using Python with Ardupilot running on Pixhawk flight computer
- Developed a program to scan QR codes using Python and OpenCV to command the UAV to fly to the QR code location

Skills

Firmware & Electronics: C/C++, STM32 & other ARM Cortex-M MCUs, RTOS, FreeRTOS, serial protocols (I2C, SPI, CAN, UART), hardware debugging (SWD, oscilloscope, logic analyzer, DMM), LTspice, Altium Designer, board bring-up & validation, soldering
Software: Python (Numpy, Pandas, etc.), Matlab, Simulink, Docker, Git & Github, Linux CLI
CAD/CAM: CSWA Certification - SolidWorks modeling, design, assembly, drawing, 3D printing